

Algorithm for 1-bit Error Detection & Correction

$$D_0, D_1, D_2, \dots, D_{N-1}, P_0, P_1, \dots, P_{r-1} \longrightarrow \boxed{\text{CHANNEL}} \longrightarrow \hat{D}_0, \hat{D}_1, \hat{D}_2, \dots, \hat{D}_{N-1}, \hat{P}_0, \hat{P}_1, \dots, \hat{P}_{r-1}$$

Using parity bits we can detect the one bit error, there will be total 2^r cases with r bits, if we give one case for no error, in total $2^r - 1$ cases, using this we can locate the error.

So, $2^r - 1 \geq N$, so that we will be able to locate the error in every bit position

Let's take example where, $r = 3$, So that $N = 4$.

$$D_0, D_1, D_2, D_3, P_0, P_1, P_2 \longrightarrow \boxed{\text{CHANNEL}} \longrightarrow \hat{D}_0, \hat{D}_1, \hat{D}_2, \hat{D}_3, \hat{P}_0, \hat{P}_1, \hat{P}_2$$

These bits can be arranged this way, $\overset{7}{D_7}, \overset{6}{D_6}, \overset{5}{D_5}, \overset{4}{D_4}, \overset{3}{D_3}, \overset{2}{D_2}, \overset{1}{D_1}$

P_1, P_2, P_4 can be related with data bits by following way:

P_4, P_2, P_1			
P_1	0	0	1
P_2	0	1	0
D_1	0	1	1
P_4	1	0	0
D_4	1	0	1
D_6	1	1	0
D_7	1	1	1

$$P_1 = D_3 \oplus D_5 \oplus D_7 \quad \& \quad P_1^+ = D_3 \oplus D_5 \oplus D_7$$

$$P_2 = D_3 \oplus D_6 \oplus D_7 \quad \& \quad P_2^+ = D_3 \oplus D_6 \oplus D_7$$

$$P_4 = D_5 \oplus D_6 \oplus D_7 \quad \& \quad P_4^+ = D_5 \oplus D_6 \oplus D_7$$

Using P^+ and $\hat{P}$ syndrome (S) can be found

$$S_0 = P_1^+ \oplus \hat{P}_1$$

$$S_1 = P_2^+ \oplus \hat{P}_2$$

$$S_2 = P_4^+ \oplus \hat{P}_4$$

Error detection can be found out using S

	S_2	S_1	S_0	
	0	0	0	$\longrightarrow$ No Error
P_1	0	0	1	$\longrightarrow$ Error In $\hat{P}_1$
P_2	0	1	0	$\longrightarrow$ Error In $\hat{P}_2$
D_1	0	1	1	$\longrightarrow$ Error In $\hat{D}_3$
P_4	1	0	0	$\longrightarrow$ Error In $\hat{P}_4$
D_4	1	0	1	$\longrightarrow$ Error In $\hat{D}_5$
D_6	1	1	0	$\longrightarrow$ Error In $\hat{D}_6$
D_7	1	1	1	$\longrightarrow$ Error In $\hat{D}_7$

Error Correction can be done by just inverting that error detected bit.

$$\Rightarrow \text{Corrected bit} = \overline{\text{Error bit}}$$